Procurement Strategy

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Topics to cover this week

1. Procurement fundamentals
2. Procurement routes
3. Tendering methods
4. Strategy selection and risk allocation
5. Group discussion



What is procurement in Civil Engineering?

The process of identifying, selecting, and acquiring the design and construction services required to deliver a project

- Set out the organizational structure in relation to the design and construction of a project
  - Participants: client, design team, contractor, subcontractors, suppliers, and management personnel
- Define the roles, responsibilities, and relationships between various project participants
- Allocate risks between the client and contractor

Project need → Procurement strategy → Tendering → Contract award → Delivery



What decisions does procurement strategy involve?

A set of decisions about responsibility, timing, competition, price formation, and uncertainty

- How should design and construction responsibilities be divided?
- When does the contractor become involved?
- How should the main project interfaces be organized?
- What type of market engagement is appropriate?

Example: LED replacement vs shading retrofit



Different but related decisions

Decision	Main question	Example
Procurement route	How are design and construction responsibilities structured?	Traditional, Design & Build
Tendering method	How will suitable contractors be identified and selected?	Open, selective, negotiated
Contract arrangement	How are obligations and commercial consequences formally governed?	Payment, variations, time obligations, defects, key risk allocation



Start with client objectives/priorities

Potential priorities factors

- Time - How important is early completion or an early start on site?
- Cost certainty - How important is knowing the final cost before construction begins?
- Design control - How much control does the client want over the detailed solution and appearance?
- Quality/performance - How important are specified technical or performance outcomes?
- Flexibility - How much ability does the client need to change the design or scope during development?



Project characteristics also matter

Project characteristics may include

- design maturity
- technical complexity
- specialist input required
- scope certainty
- occupied-site constraints
- project size
- repetition / standardization
- client capability / experience



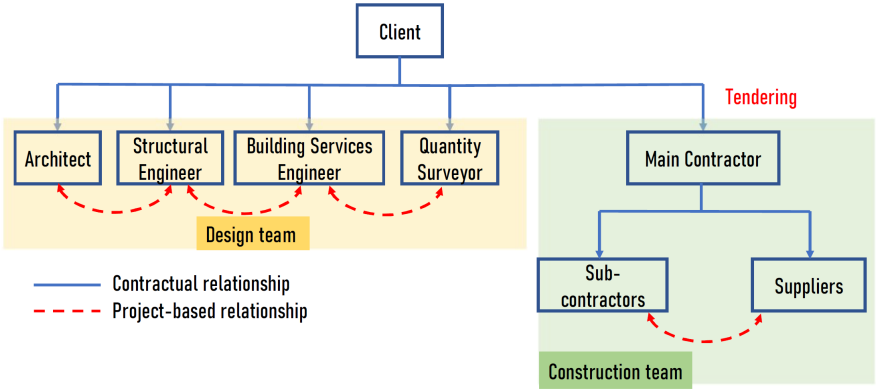
Same project, different delivery structure

The project scope may be the same, but procurement changes who is responsible for delivering it and how the parties relate to each other.

Activity	Possible responsible party
Define project requirements	Client
Develop detailed design	Client's consultant or contractor
Construct/install	Contractor
Coordinate design and construction	Client/consultant or contractor
Test and demonstrate performance	Contractor, verified by client/consultant



Traditional approach



Design and construction contractually separated

Stage 1: Client (appoints an architect/consultant to) come up with the design before entering into a contract with a contractor.

- Design substantially developed **before** contractor appointment
- Contractor tenders based on the (mostly) completed design

Stage 2: Client enters into a contract with the selected contractor.

- Contractor is mainly responsible for construction
- Designer remains engaged during construction for administration/inspection



Traditional procurement in practice

In the LED replacement project, client/consultant defines:

- lighting type/level requirements
- locations/quantities
- control requirements
- installation constraints

Then the contractor:

- prices the defined works
- supplies and installs
- tests and hands over



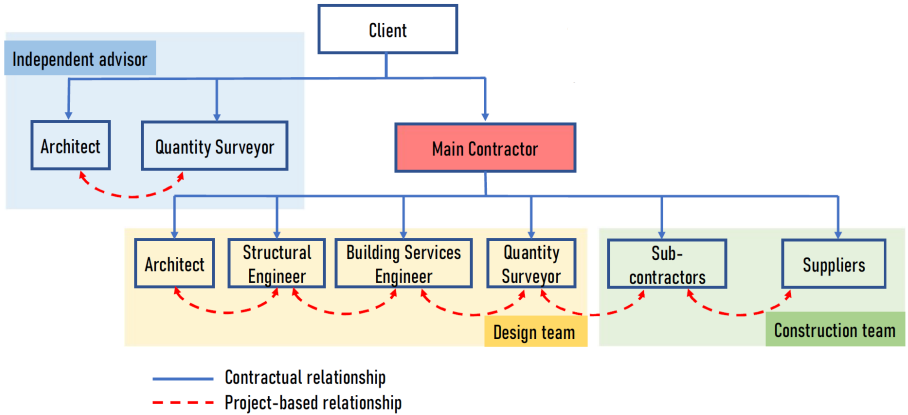
Pros and Cons

- ✓ Strong client control over design
- ✓ Clear scope and high certainty in price
- ✓ Easier evaluation of bids against the same design

- × Longer project duration with separated design and construction
- × Contractors' inputs regarding feasibility come relatively late
- × Project management after tender can be more challenging



Design and Build



Integrated responsibility

Client

- Provide a clear concept design and performance specifications
- Can be provided by independent advisors
- Examine the detailed design produced by the contractor

Contractor

- Assumes responsibility for detailed design and construction
- May employ its own designers or appoint consultants
- Client deals primarily with one main contractor



Design and Build in practice

In the shading retrofit project, client/consultant defines:

- target reduction in solar heat gain
- comfort requirements
- architectural constraints
- control functionality
- required performance

Then the contractor develops:

- shading configuration
- actuator system
- control logic
- detailed interfaces
- installation method

before construction.



Pros and Cons

- ✓ Strong design-construction integration
- ✓ Potentially shorter overall program
- ✓ Clearer single-point responsibility
- ✓ Majority of the risks passed on to the contractor

- × Less control over detailed design
- × Harder to compare different technical solutions
- × Harder to track and react to variations and changes



Key difference: who controls the design?

	Traditional	**Design & Build**
Who appoints the designer?	Client	Contractor
Who develops detailed design?	Client's consultant	Contractor's team
When does contractor enter?	After substantial design	Earlier
Who coordinates in construction?	Client/consultant-led	More contractor-led

Design, Novate, and Build: the designer's appointment is initiated with the client and later transferred to the contractor.

- The boundary between Traditional and D&B is not always absolute.



Other procurement routes

Framework/term contracts

- Client has repeated similar works
- Contractor capability has already been established
- Individual work packages are called off as needed

Management contracting

- Useful for large or complex projects
- Management contractor advises on feasibility and sequencing during design
- Divide construction into multiple packages for procurement and management



Tendering methods

Tendering method determines how the client approaches the market and selects contractors to compete for the work.

- The same procurement route can use different tendering methods.



Open tendering

Any contractor meeting the stated requirements may submit a tender.

Public invitation → Tender submissions → Evaluation → Award

- Broadest market access
- Transparent opportunity to participate
- Potentially large number of tenders
- High level of competition
- Higher burden for tender evaluation (cost)

e.g. Relatively standard project with many capable local contractors



Selective tendering

Only contractors that have been shortlisted or prequalified are invited to tender.

Prequalification/shortlist → Invitation → Tender submissions → Evaluation → Award

- Stronger confidence in bidder capability
- More straightforward tender evaluation
- Fewer bidders and less market competition
- Weaker public accountability, potential unethical issues

e.g. Complex project requiring relevant experience from contractors



Prequalification

Screening contractors before inviting them to submit full tenders.

- Relevant project experience
- Technical capability
- Financial capacity
- Safety record
- Staffing and resources
- Required licenses/certifications

Open tender with qualification requirements V.S. Selective tendering following prequalification



Non-competitive tendering

Negotiated Tendering: The client negotiates directly with one contractor, or occasionally a very small number of contractors, instead of running broad competition.

- Limited competition, therefore weaker market price comparison
- Potentially faster procurement decision
- Continuity or specialist knowledge required
- Clear justification usually needed

Possible situations: urgent works (e.g. emergency repair), highly specialist contractor, continuation of existing work (*Serial Tendering*), compatibility with existing systems.



Which tendering method fits your project?

	Open	Selective	Negotiated
Competition	High	Moderate–high	Low
Bidder capability control	Lower initially	High	High
Evaluation burden	High	Moderate	Low
Tendering/admin cost	High	Moderate	Low
Price tension	High	Usually good	Weaker
Speed	Usually slower	Moderate	Potentially faster

The tendering method should provide sufficient competition and capability assurance without imposing disproportionate procurement cost.

Given your project scope, complexity, and required contractor capability, who should be invited to tender, and why?

Strategy selection - identify the dominant project drivers

The criteria are not equally important for different projects.

- completion deadline
- cost certainty
- importance of control in design
- complexity of interfaces
- need for specialty
- likelihood of scope changes
- contractor market capability

Example: Campus HVAC controls retrofit



Strategy selection - separate requirements from preferences

Hard constraints

- must finish before semester starts
- contractor must hold specific certification
- works must remain compatible with existing system
- procurement value within a certain approval threshold

Preferences

- stronger cost certainty
- broader competition
- earlier contractor involvement

All options → remove infeasible options → compare remaining options



Strategy selection - evaluate the combined strategy

Does the strategy work as a package?

D&B + prescriptive design
→ unclear design responsibility

Incomplete design + fixed lump sum
→ price certainty without scope certainty

Specialist project + unrestricted tendering
→ market approach does not reflect capability requirements

A procurement strategy should be internally consistent: responsibility, competition, pricing, and project uncertainty need to support one another.



Procurement changes responsibility, not uncertainty

Procurement changes who manages and bears particular uncertainties, but it doesn't make uncertainties disappear.

- Traditional procurement → more design-interface risk remains around the client/consultant side
- D&B → more design-performance risk moves toward the contractor
- Open tendering → unsuitable bidders more likely to reach the evaluation stage
- Framework → lower repeated procurement effort, but possible dependence on a limited contractor pool

What risks does this strategy create, retain, reduce, or shift?



Principles of risk allocation

Allocate each risk to the party best able to manage it at a reasonable cost.

- Who has the best information?
- Who can influence whether it occurs?
- Who can mitigate its consequences?
- Who can price it most efficiently?

E.g. campus access restrictions, contractor productivity, unknown concealed services



Integrated example - campus water-efficiency retrofit

Project: HKU plans to upgrade washroom fixtures, install leak sensors, and improve water metering across multiple campus buildings over the next three years.

- many small, repetitive retrofit packages
- exact buildings and quantities are not all known at the beginning
- technical requirements are relatively standardized
- disruption to occupied buildings must be minimized
- the university wants consistent quality and pricing across projects



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Recommended strategy: framework/term contract + selective tendering + call-offs based on pre-agreed rates



Integrated example - potential risks

Risk	When it matters
Future quantities differ from estimates	Total programme scope is uncertain
Contractor capacity	Several call-offs may arise simultaneously
Rate escalation	Framework may run for several years
Work in occupied buildings	Work must fit campus operations
Existing plumbing conditions	Actual conditions may differ by building
Performance of installed fixtures	Water savings depend on product and installation quality

Group discussion

Project selection and planning

